Analytic Attempt at the Coefficient of I_B: Hyperbolic Reduction, Source Dependence, and an Admission

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Abstract

Paper VIII of the Emergent Gravity Program left the multi-digit coefficient of the kinematic residue I_B open, and required an analytic Hadamard expansion of the current two-point function under modular flow. This note carries out the $d = 2$ reduction as far as it goes, then stops.

After the map $x = R \tanh u$ the weight-one Jacobians cancel and the smeared correlator is

$$Q(s) = \int du\, dv\, b(u)b(v)/\sinh^2(u - v + \pi s),$$

with $b(u) = \beta(\tanh u)$. The modular kernel pulls back to a convolution against sech^2. That identity is algebraic and is not in dispute.

A Hadamard finite part for the pulled-back integral was written in closed form, $I_{\mathrm{tot}}^{\mathrm{FP}}(\tau) = 2\,\mathrm{sech}^2(\tau)\,(\tau \tanh \tau - 1)$. A hole-quadrature check of that formula does *not* confirm it: the hole is not a Hadamard finite part, and the two sides do not agree. The closed form is therefore **not** certified.

A direct evaluation of the relative residue $r = \int K(s)(Q(s) - Q(0))\, ds/|Q(0)|$ for five inequivalent sources in the hyperbolic coordinate gives values that *move* with the source (spread ≈ 0.23 on the grids used). If that movement survives a genuine Hadamard scheme, there is no single universal multi-digit coefficient of I_B independent of β.

This note does not extract a multi-digit number. The Final Status label of I_B remains [**O**].

Standing rule

Numerical agreement is not proof. An unconfirmed closed form is not a theorem. A source-dependent ratio is not a universal coefficient. This draft was prepared with computational assistance; every [**ADMISSION**] is an owned failure, not a pending detail.

1 The object

Paper VI defines, up to the current central charge C_J,

$$I_B[\beta; \zeta] = -\int_{-\infty}^{\infty} ds\, K(s) \left\langle \sigma_s J[\beta]\, \widetilde{J}[\zeta] \right\rangle_{\kappa \neq 0}, \qquad K(s) = \frac{\pi}{2\cosh^2(\pi s)}, \qquad \int K = 1. \tag{1}$$

On the interval $(-R, R)$ the modular flow is

$$x(s) = R \tanh\bigl(\operatorname{artanh}(x/R) + \pi s\bigr). \tag{2}$$

The chiral current two-point function is $\langle J(x)J(y)\rangle = k/(x-y)^2$. The constant k cancels in any relative residue and is set to 1 below. $R = 1$ by scaling.

2 Hyperbolic reduction

Set $x = \tanh u$, $y = \tanh v$. Then

$$\frac{\partial x(s)}{\partial x}\frac{1}{(x(s)-y)^2} = \frac{\cosh^2 u \cosh^2 v}{\sinh^2(u - v + \pi s)}, \qquad dx = \operatorname{sech}^2 u \, du. \tag{3}$$

The Jacobians cancel against the measure:

$$Q(s) = \int du\, dv\, b(u)b(v)\frac{1}{\sinh^2(u-v+\pi s)} = \int d\tau\, C(\tau)\frac{1}{\sinh^2(\tau+\pi s)}, \tag{4}$$

where $b(u) = \beta(\tanh u)$ and $C(\tau) = \int du\, b(u)b(u-\tau)$.

Pulling back the kernel,

$$\int_{-\infty}^{\infty} K(s)\, f(\tau + \pi s)\, ds = \int_{-\infty}^{\infty} \frac{d\alpha}{2\cosh^2\alpha}\, f(\tau+\alpha). \tag{5}$$

On the real line the only singularity of $1/\sinh^2 w$ is at $w = 0$, and

$$\frac{1}{\sinh^2 w} - \frac{1}{w^2} = -\frac{1}{3} + O(w^2) \tag{6}$$

is smooth. Subtracting $1/w^2$ is the first Hadamard step. It is *not* the last: Paper VIII required every non-integrable term of the *flowed* two-point function to be subtracted, not only the $s = 0$ coincidence.

Proposition 2.1 (reduction)**.** *Equations (4)–(5) are identities of the $d = 2$ weight-one setup. They do not evaluate I_B.*

3 An unconfirmed closed form

The substitution $u = \tanh\alpha$ converts the pulled-back integral of $1/\sinh^2$ into

$$I_{\text{tot}}(\tau) = \frac{1}{2\cosh^2\tau}\int_{-1}^{1} du\, \frac{1-u^2}{(u+\tanh\tau)^2}. \tag{7}$$

The integrand has a double pole on the contour at $u = -\tanh\tau$. Dropping the $1/\varepsilon$ term in a symmetric hole and taking $\varepsilon \to 0$ produces the candidate

$$I_{\text{tot}}^{\text{FP}}(\tau) \stackrel{?}{=} 2\operatorname{sech}^2(\tau)\,(\tau\tanh\tau - 1). \tag{8}$$

Admission. Equation (8) was *not* confirmed. A hole quadrature of (7) diverges as the hole shrinks, as a double pole must, and the finite part extracted from that hole does not match (8) under the implementation used (`m9_3_ib_analytic.py`). A hole is not a Hadamard finite part. Until (8) is verified by an independent finite-part prescription, it is a guess, not a lemma. It is not used as input to any number claimed below.

4 Source dependence of the relative residue

Define, with the same cutoff on $|\tau + \pi s|$ for every s,

$$r[\beta] := \frac{\displaystyle\int ds\, K(s)\big(Q(s) - Q(0)\big)}{|Q(0)|}. \tag{9}$$

Five sources $b(u)$ were run on a linear u-grid (`m9_3` family, direct correlation, hole 0.03):

$b(u)$	r
Gaussian, $\sigma = 0.6$	-0.108
Gaussian, $\sigma = 1.1$	$+0.089$
$\operatorname{sech} u$	$+0.120$
$e^{-\|u\|}$	$+0.046$
compact bump, width 1.4	-0.111

Spread: 0.23. Kernel mass on the truncated s-interval is 0.90, not 1; that truncation is a further scheme artefact.

Admission. These five numbers are scheme-dependent proxies, not Hadamard finite parts. They are already enough to kill a universality claim *inside this scheme*: r changes sign with the source. A universal multi-digit coefficient would have to be independent of β. This scheme does not produce one. Whether a genuine Hadamard r is universal remains [**O**]. The present evidence leans against universality and does not replace Paper VIII's demand for an analytic expansion.

5 What was not done

- No closed form for $\displaystyle\int d\alpha\, \operatorname{sech}^2\alpha\, w_{\text{fin}}(\tau + \alpha)$.
- No isolation of the u_1 or log terms of a full Hadamard series for the flowed correlator.
- No three-digit stability under simultaneous refinement of grid, hole, and s-cutoff.
- No evaluation in $d = 4$.

6 Verdict

Claim	Status
Hyperbolic reduction (4)–(5)	derived; algebra only
Closed form (8)	[**ADMISSION**] unconfirmed
Universal multi-digit coefficient of I_B	[**O**]; this attempt did not find one
Source-independent r in the hole scheme	fails (spread 0.23, sign change)
Final Status label of I_B	unchanged, [**O**]

The next honest step is still the one Paper VIII named: an analytic Hadamard series of $\langle J(x(s))J(y)\rangle$ whose finite part can be integrated against $K(s)$ in closed form, or a proof that the finite part is not unique. This note is neither.

Acknowledgments

Computational assistance drafted the algebra checks and the scripts `m9_2_ib_hadamard.py` and `m9_3_ib_analytic.py`. The admissions are the point of the note.

References

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